CM0845 Logic
First-Order Logic: Natural Deduction

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Preliminaries

Textbook

D. van Dalen (2013). Logic and Structure. Sections 3.8 and 3.9.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Derivation Rules for the Quantifiers

Universal quantifier

$$\frac{\varphi(x)}{\forall x \varphi(x)}\,\forall\mathrm{I} \qquad\qquad \frac{\forall x \varphi(x)}{\varphi[\,t/x\,]}\,\forall\mathrm{E}$$

Side condition: In $\forall\mathrm{I}$, the variable x may not occur free in any hypothesis on which $\varphi(x)$ depends.

Existential quantifier

$$\frac{\varphi[\,t/x\,]}{\exists x \varphi(x)}\ \exists\mathrm{I}$$

$$\frac{\exists x \varphi(x) \qquad \begin{array}{c}[\varphi(x)]^x \\ \vdots \\ \psi\end{array}}{\psi}\ \exists\mathrm{E}^x$$

Side condition: In $\exists\mathrm{E}$, the variable x is not free in ψ, or in a hypothesis of the sub-derivation of ψ, other than $\varphi(x)$.

Derivation Rules for the Quantifiers

Example

The derivation rules for the quantifiers are consistent with the convention that the universe of discourse is not empty, so we can prove that $\vdash \forall x \varphi(x) \to \exists x \varphi(x)$.

Proof

$$\dfrac{\dfrac{\dfrac{[\forall x \varphi(x)]^x}{\varphi(x)}\,\forall\mathrm{E}}{\exists x \varphi(x)}\,\exists\mathrm{I}}{\forall x \varphi(x) \to \exists x \varphi(x)}\,{\to}\mathrm{I}^x$$

Example

Prove that $\vdash \exists x(\varphi(x) \vee \psi(x)) \to \exists x\varphi(x) \vee \exists x\psi(x)$ (van Dalen 2013, p. 92).

Proof

$$
\dfrac{\dfrac{[\exists x(\varphi(x) \vee \psi(x))]^w \qquad \dfrac{[\varphi(x) \vee \psi(x)]^z \qquad \dfrac{\dfrac{[\varphi(x)]^x}{\exists x\varphi(x)}\,\exists\mathrm{I}}{\exists x\varphi(x) \vee \exists x\psi(x)}\,\vee\mathrm{I} \qquad \dfrac{\dfrac{[\psi(x)]^y}{\exists x\psi(x)}\,\exists\mathrm{I}}{\exists x\varphi(x) \vee \exists x\psi(x)}\,\vee\mathrm{I}}{\exists x\varphi(x) \vee \exists x\psi(x)}\,\vee\mathrm{E}^{x,y}}{\exists x\varphi(x) \vee \exists x\psi(x)}\,\exists\mathrm{E}^{z}}{\exists x(\varphi(x) \vee \psi(x)) \to \exists x\varphi(x) \vee \exists x\psi(x)}\,{\to}\mathrm{I}^{w}
$$

Derivation Rules for the Identity

Identity

$$\frac{}{x = x}\,\mathrm{RI}_1 \qquad \frac{x = y}{y = x}\,\mathrm{RI}_2 \qquad \frac{x = y \qquad y = z}{x = z}\,\mathrm{RI}_3$$

$$\frac{x_1 = y_1, \ldots, x_n = y_n}{t[\, x_1, \ldots, x_n / z_1, \ldots, z_n \,] = t[\, y_1, \ldots, y_n / z_1, \ldots, z_n \,]}\,\mathrm{RI}_4$$

$$\frac{x_1 = y_1, \ldots, x_n = y_n \qquad \varphi[\, x_1, \ldots, x_n / z_1, \ldots, z_n \,]}{\varphi[\, y_1, \ldots, y_n / z_1, \ldots, z_n \,]}\,\mathrm{RI}_4$$

Derivation Rules for the Identity

Example

The derivation rules for the identity are consistent with the convention that the universe of discourse is not empty, so we can prove that $\vdash \exists x(x = x)$.

Proof

$$\dfrac{\dfrac{}{x = x}\ \mathrm{RI}_1}{\exists x(x = x)}\ \exists\mathrm{I}$$

References

Dirk van Dalen [1980] (2013). Logic and Structure. 5th ed. Springer (cit. on pp. 2, 6).